

1(a)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Year": range(2015, 2025),
    "Postings": [1200, 1500, 1800, 2300, 2800, 2500, 4000, 5200, 6100, 7000]
})

print(df)
print("\nGrowth:", ((df.Postings.iloc[-1] / df.Postings.iloc[0]) - 1) * 100, "%")

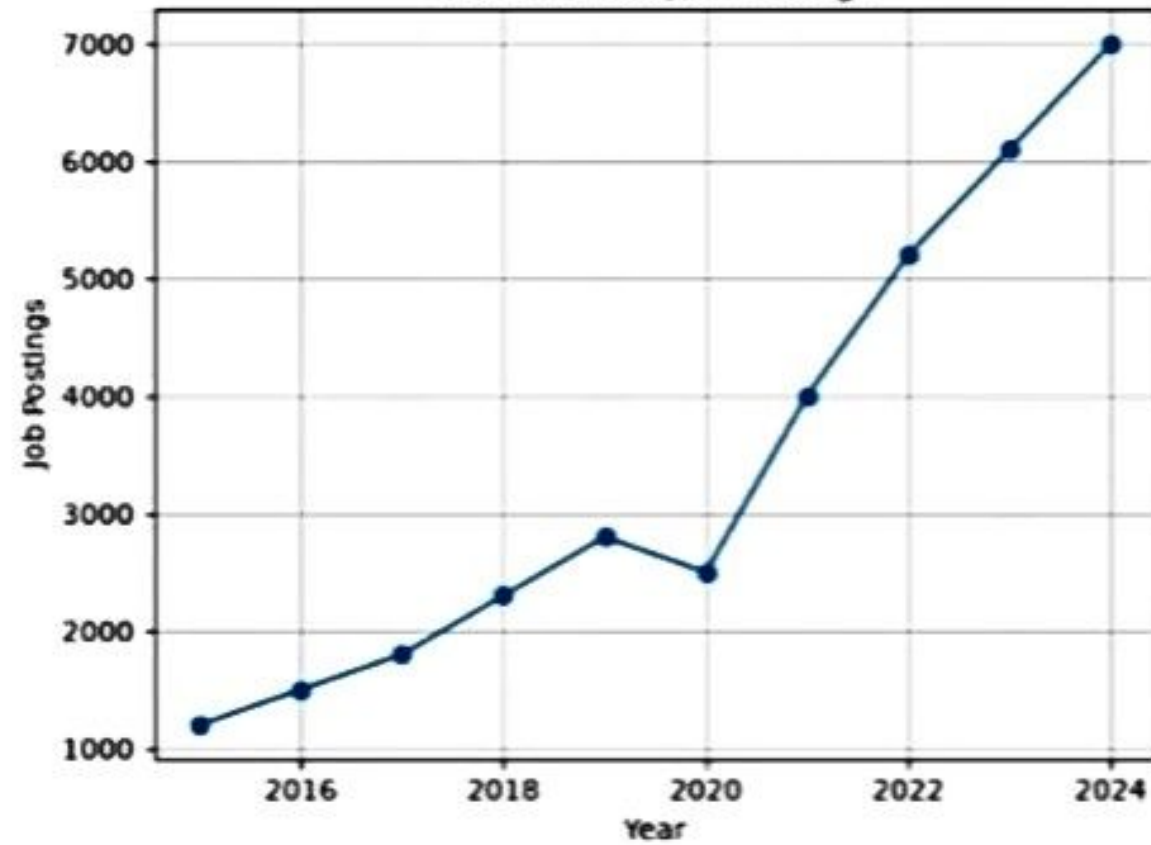
plt.plot(df.Year, df.Postings, marker="o")
plt.title("Data Science Job Postings")
plt.xlabel("Year")
plt.ylabel("Job Postings")
plt.grid()
plt.show()
```

plt.show()

	Year	Postings
0	2015	1200
1	2016	1500
2	2017	1800
3	2018	2300
4	2019	2800
5	2020	2500
6	2021	4000
7	2022	5200
8	2023	6100
9	2024	7000

Growth: 483.3333333333333 %

Data Science Job Postings



1(b)

```
[3]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Role": ["Data Analyst", "Data Engineer", "Data Scientist",
            "ML Engineer", "Business Analyst"],
    "Count": [250, 180, 220, 150, 120]
})

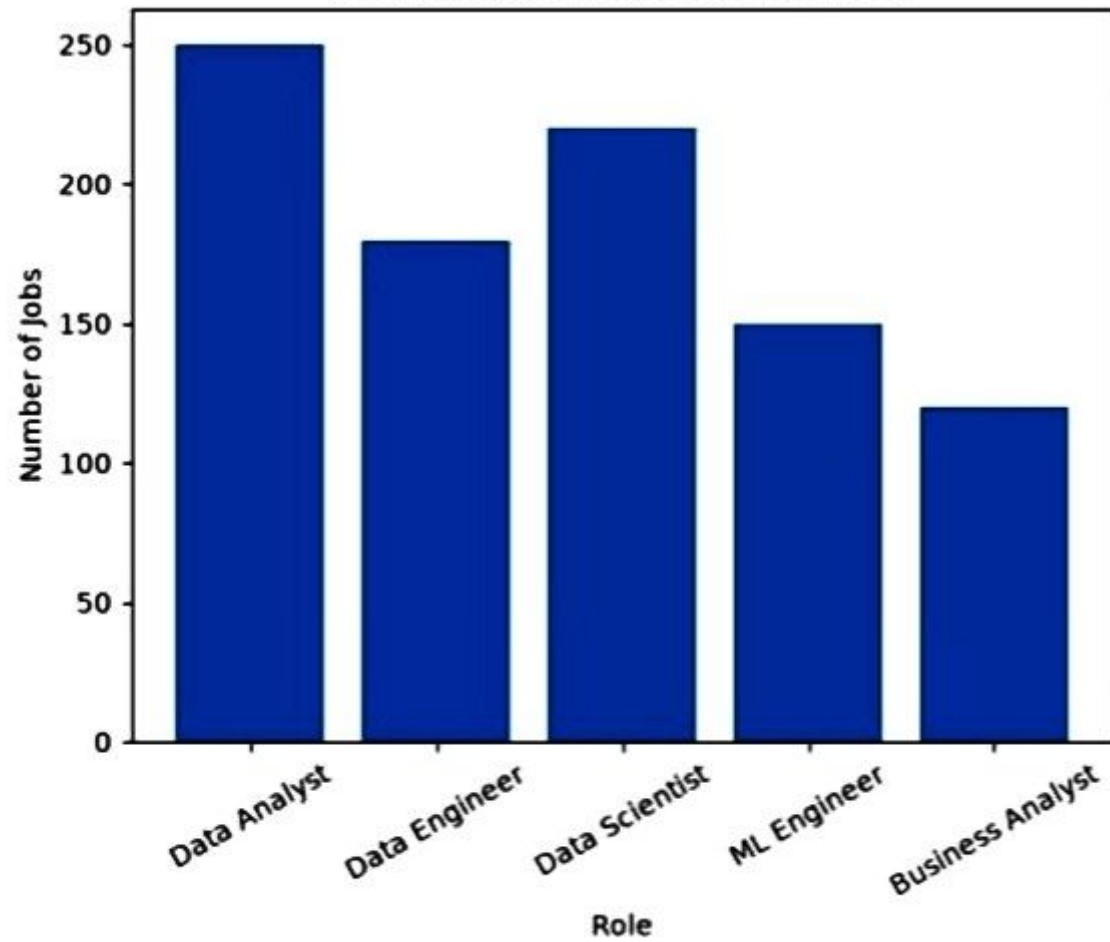
print(df)

plt.bar(df["Role"], df["Count"])
plt.title("Distribution of Data Science Roles")
plt.xlabel("Role")
plt.ylabel("Number of Jobs")
plt.xticks(rotation=30)
plt.show()
```

```
plt.show()
```

	Role	Count
0	Data Analyst	250
1	Data Engineer	180
2	Data Scientist	220
3	ML Engineer	150
4	Business Analyst	120

Distribution of Data Science Roles



1(c)

```
import pandas as pd

structured = pd.DataFrame({"Name": ["A", "B"], "Age": [20, 21]})
semi_structured = [{"Name": "A", "Age": 20}, {"Name": "B", "Age": 21}]
unstructured = ["Hello world", "Data science"]

print("Structured:\n", structured)
print("\nSemi-Structured:\n", semi_structured)
print("\nUnstructured:\n", unstructured)
```

Structured:

	Name	Age
0	A	20
1	B	21

Semi-Structured:

```
[{'Name': 'A', 'Age': 20}, {'Name': 'B', 'Age': 21}]
```

Unstructured:

```
['Hello world', 'Data science']
```

1(d)

```
5]: from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
```

```
f = Fernet(key)
```

```
data = "Sensitive Data"
```

```
encrypted = f.encrypt(data.encode())
```

```
decrypted = f.decrypt(encrypted).decode()
```

```
print("Encrypted:", encrypted)
```

```
print("Decrypted:", decrypted)
```

```
Encrypted: b'gAAAAABqezZd8WfW6ny-yODTdAct4-p0J3b6D0Df10ER9yPrTVeqndTpJkrRQpK5qmx-v4uKfMjBCAwzAX-KMITVMsLjsRf09w=='
```

```
Decrypted: Sensitive Data
```